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# Crypto freedom! Effects of trait reactance and regulation content on intention to buy cryptocurrency

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ABSTRACT

Cryptocurrencies like Bitcoin have attracted billions in investment in a short period of time. Currently, there is a move to introduce regulations for crypto trading. The purpose of the current research was to study the effect of trait reactance and messages about freedom-protecting (vs. freedom-restricting) crypto regulations on an individual's attitude toward cryptocurrency and buying intention. The following mediators were examined: (1) positive affect and (2) negative affect (i.e., anger). A total of 566 people took part in a 2 (regulation content: freedom-protecting vs. freedom-restricting) × trait reactance (continuous) between-participants design. The freedom message emphasized that regulations would support an individual investor's right to invest in crypto. The restriction message stated that crypto trading would be restricted to professional investors. Trait reactance was measured with Hong and Page's (1989) Psychological Reactance scale. Results indicated that trait reactance had a positive effect on state reactance and positive affect. The effect of trait reactance on crypto buying intention was mediated by positive affect and anger. Freedom (vs. restriction) messages were associated with an increase in positive affect and decrease in anger. Freedom messages about crypto investing generate a more positive response than messages outlining restrictions to crypto investing.

## 1. Introduction

Investor interest in cryptocurrency ("crypto") has increased in recent years. The global crypto market has changed from USD$190 billion to over $1.7 trillion US dollars in the last two years (Coinmarketcap.com, 2022). Crypto is a digital asset that exists on a blockchain and which can be transferred without using an intermediary (e.g., a bank). A consequence of cryptocurrency trading being independent of government intermediaries is that investors have no redress if they lose their money to fraud. With a large and volatile crypto market, a lack of regulations protecting investors and/or lack of government control over this area, news is emerging that regulations will be introduced for crypto trading. Indeed, crypto regulation is emerging in the US, India, Australia, Russia, and Botswana (Bain, 2022; Erazo, 2022; Eyers, 2022; Rearick, 2022; Singh, 2022).

Public discourses on crypto regulation tend to focus on one of two aspects. On one hand, regulators have emphasized protecting investors. For instance, the Chair of the U.S. Securities and Exchange Commission (SEC), Gary Gensler has advocated crypto investor protection (Bain, 2022) and the SEC is acting to ensure crypto financial services conform to security regulations (U.S. Securities and Exchange Commission, 2022). Similarly, when the popular crypto exchange Binance was granted a virtual asset licence in Dubai, the announcement highlighted the regulatory framework that would be apply (Reuters, 2022). Yet discussions also revolve around proposals that restrict investor freedom. For example, in Hong Kong which houses some of the world's largest crypto exchanges, the Financial Services and the Treasury Bureau (FSTB) announced that it would restrict crypto trading to professional investors only (defined as having a portfolio of approximately $1 million, John, 2021). Relatedly, many opinion-based websites and online discussions discuss crypto and taxation, either in relation to saving money or to avoid government oversight of crypto assets (Cothern, 2022). Tax is also relevant as a crypto tax has been signalled or introduced by nations (Singh, 2022).

The present research considers potential crypto regulation in terms of an individual's trait reactance and whether crypto regulations protect (vs. restrict) an individual's freedom to buy crypto. Psychological reactance theory (Brehm, 1966; Brehm & Brehm, 1981) predicts that when

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something removes or threatens an individual's freedom of behavior, a person will experience psychological reactance which is a motivational state to regain that freedom (Rosenberg & Siegel, 2018). Decades of psychological research shows that reactance results in cognitions and behaviors to re-establish one's freedom (e.g., engaging in the prohibited behavior; threatening a person to remove the restriction) as well as negative affect (e.g., anger) at the loss of freedom (see Miron & Brehm, 2006; Rosenberg & Siegel, 2018; Steindl et al., 2015 for reviews). Trait reactance is an individual difference in reactance proneness (Shen & Dillard, 2005) which has been shown to offer insight into how people react to directives. For example, in a medical context, trait reactance was associated with greater resistance to anti-smoking advice received from physicians (Graybar et al., 1989). Regarding COVID-19, trait reactance was negatively associated with perceptions of physicians' communication about vaccination directives and whether an individual prioritizes vaccination (Finkelstein et al., 2020). In a multinational sample of 5323 participants trait reactance was positively associated with anti-vaccination attitudes (Hornsey et al., 2018).

For consumers, trait reactance influences consumer responses to retail promotions that require consumers to perform certain actions to receive a discount (Bertini & Aydinli, 2020). In the present research, we examine whether trait reactance interacts with messages about crypto regulations that protect or restrict investor freedom. Regulations that restrict an investor's freedom to invest in crypto may be viewed particularly negatively by people who are higher in trait reactance. This is relevant as reactance appears to be present in the population. Campbell (2020) found in a survey of 3839 people across the U.S. that psychological reactance was normally distributed with a mean of 3.36 ($SD$ = 0.92) on a 0–6 scale. Similarly, LaVoie et al. (2017) found moderate levels of trait reactance ($M$ = 3.87, $SD$ = 0.80) from a sample of 435 Midwestern undergraduates.

We also investigate an area that has been less examined in reactance research – when high reactance people respond *positively* to messages. While a person may experience anger from restricted freedom, what happens if someone high in trait reactance is exposed to a message that shows that their freedoms are protected? We also examine this issue. Thus, the present research studies the effect of trait reactance and regulation content (freedom protecting vs. freedom restricting) on crypto attitude and buying intention. Two mediators that may drive these effects are studied: positive affect and anger.

### 1.1. The present study

In the present research, we explored whether trait reactance and regulation content influenced crypto judgments. We expected trait reactance to result in a positive effect on a person's response to freedom-protecting messages and a negative effect for restriction messages. Higher-reactance individuals exposed to a freedom (vs. restriction) message should exhibit a more favorable attitude toward cryptocurrency and buying intention, than individuals who are lower in reactance. We also explored the mediating role of affect in individual responses to messages about crypto regulations with both positive and negative affective pathways (Fig. 1).

For positive affect, insight can be gained from work in message framing which has found that positive messages that support a person's important goals create positive affect. In contrast, negative, goal-incongruent messages can create negative affect. For example, Shen and Dillard (2007) in a study of health messages found that advantage (vs. disadvantage) framing resulted in stronger positive (vs. negative) emotions.

Regarding crypto buying intention, prior research on individual differences and consumer cryptocurrency buying intentions has found that positivity can drive crypto buying intention. Martin et al. (2022) found that the effect of narcissism on crypto judgments was mediated by positivity. Likewise, we predict that high-trait reactance people are likely to experience positive affect when exposed to a message about protecting their freedom to invest in crypto. Thus, we predict that the effect of trait reactance and freedom-protecting regulations on crypto judgments is mediated by positive affect.

In contrast, when people are exposed to messages about restrictions to their freedom to buy crypto, then a negative affective response is possible. We predict that messages about restrictive regulations should generate negative affect from high-trait reactance people, specifically, anger. Anger is a component of state reactance along with negative cognitions (Dillard & Shen, 2005; Rains, 2013). Steindl et al. (2015) in a review of reactance research suggest that anger is an important component of state reactance and that people who have their freedom threatened usually feel angry. It is plausible that reactance prone people are likely to experience anger when exposed to freedom-restricting messages. Thus, we predict that the effect of trait reactance and freedom-restricting messages on crypto judgments is mediated by anger.

## 2. Method

### 2.1. Participants and procedure

Participants were drawn from Prolific Academic. The study was approved by the Ethical Committee of Cardiff University (Number 2122018) and started in November 2021. Data collection took place in two waves. In the first wave ($N$ = 799; females = 49.3%, other/prefer not to say = 0.6%; $M_{age}$ = 39.70; SD = 13.4) participants reported their awareness of cryptocurrency and their interest in investing in financial assets. Participants aware about cryptocurrency, interested in investing, and agreeing to participate in a follow up survey ($N$ = 588) were invited for the main study ($N$ = 566; females = 44.2%, other/prefer not to say = 0.7%; $M_{age}$ = 39.10 years; $SD$ = 13.00; return rate = 96.3%). Table 1

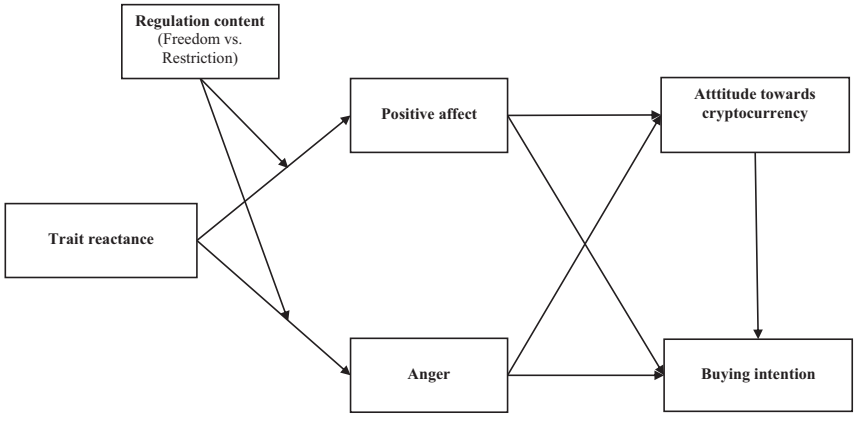



**Fig. 1.** Conceptual model.

**Table 1**
Sample background ($N = 566$).

| | N (%) | | N (%) |
|---|---|---|---|
| Gender (%) | | Are you aware of cryptocurrency? (%) | |
| Male | 312 (55.1) | Yes | 566 (100.0) |
| Female | 250 (44.2) | No | – |
| Non-binary/third gender | 3 (0.5) | Are you interested in investing in stocks, bonds, or cryptocurrency? (%) | |
| I do not want to answer | 1 (0.2) | Yes | 316 (55.8) |
| Mean age (SD) | 39.10 (12.96) | Maybe | 250 (44.2) |
| Education (%) | | No | – |
| Primary school | 2 (0.4) | What is your experience in investing in stocks, bonds, or cryptocurrency? (%) | |
| Post primary school | 39 (6.9) | Never | 176 (31.1) |
| Further education | 177 (31.3) | Less than a year | 148 (26.1) |
| Undergraduate higher education | 241 (42.6) | 1–5 years | 139 (24.6) |
| Professional education | 97 (17.1) | 6–10 years | 38 (6.7) |
| Doctorate | 10 (1.8) | 11–15 years | 16 (2.8) |
| Marital status (%) | | 16–20 years | 13 (2.3) |
| Single | 215 (38.0) | More than 20 years | 36 (6.4) |
| Married | 209 (36.9) | Do you own any cryptocurrency? (%) | |
| Cohabiting | 116 (20.5) | Yes | 170 (30.0) |
| Divorced | 24 (4.2) | No | 396 (70.0) |
| Widowed | 2 (0.4) | [If no] Would you be interested in investing in cryptocurrency? (%) | |
| Annual Income (%) | | Yes | 212 (53.5) |
| Less than £10,000 | 99 (17.5) | No | 184 (46.5) |
| £10,000 - £19,999 | 115 (20.3) | | |
| £20,000 - £29,999 | 143 (25.3) | | |
| £30,000 - £39,999 | 91 (16.1) | | |
| £40,000 - £49,999 | 58 (10.2) | | |
| More than £50,000 | 60 (10.6) | | |

displays final sample characteristics. The Supplementary Material provides descriptive statistics for both waves and differences between groups of participants who own and are interested in investing in cryptocurrency.

The sample size was checked in terms of statistical power for model testing. A priori power analyses using G*Power was used to determine the adequacy of the sample size (Faul et al., 2009). Using suggested minimum values by Cohen (1988, a minimum $R^2$ value of 0.10, a statistical power of 80%, and 4 predictors), the a priori G*Power calculation indicated that a sample size of minimum 125 would be required. However, to allow for exploring the data (according to our pre-registered protocol) we aimed at a higher sample.

### 2.2. Measures

This section presents the measures used in the study. Internal consistency and means are presented in Table 2, scale items and descriptive measures appear in the Supplementary Material.

**Table 2**
Descriptive statistics and Pearson correlations between variables.*

| | α | Mean (SD) | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|
| 1. Attitude | 0.96 | 4.29 (1.49) | | | | |
| 2. Buying Intention | 0.97 | 4.03 (1.80) | 0.84** | | | |
| 3. Positive Affect | 0.93 | 2.49 (0.90) | 0.47** | 0.52** | | |
| 4. Anger | 0.95 | 1.52 (0.91) | 0.11** | 0.15** | −0.01 | |
| 5. Trait Reactance | 0.88 | 3.71 (0.84) | 0.17** | 0.20** | 0.14** | 0.07 |

Notes: All variables are measured on a 7-point scale, except for positive affect and anger (5-point).
** $p < 0.01$.
* $p < 0.05$.

#### 2.2.1. Dependent measures

Attitude toward cryptocurrency was assessed with three items (bad/good, unfavorable/favorable, and negative/positive) adapted from Martin et al. (2019). Buying intention was assessed with three items (unlikely/likely, unfavorable/favorable, and negative/positive) adapted from Martin and Strong (2016). All items were measured on a 7-point bipolar scale.

#### 2.2.2. Independent measures

We assessed trait reactance based on the Hong and Page (1989) scale. The scale consists of 14 items measured on a 7-point agreement scale (1 = strongly disagree; 7 = strongly agree). We manipulated regulation content by instructing participants to read a message about cryptocurrency investing. The message was freedom-protecting or freedom-restrictive with participants randomly assigned to one of the two conditions. We further conducted a manipulation check of the regulation content manipulation with US participants recruited through MTurk ($N = 152$; 46.1% females, 1.3% prefer not to say/other; $M_{age} = 41.40$ years; SD = 12.10). We assessed threat to freedom based on a 4-item scale adapted from Kavvouris et al. (2020) and Richards et al. (2020) measured on a 7-point agreement scale (1 = strongly disagree; 7 = strongly agree; $\alpha = 0.94$). Individuals in the freedom condition perceived significantly less threat to freedom (M = 2.64, SD = 1.39) compared to those in the restriction condition (M = 4.18, SD = 1.72), $t(150) = 6.06$, $p < 0.001$, *Cohen's d* = 0.984). The two messages appear in the Supplementary Material together with the manipulation check results.

#### 2.2.3. Mediators

Positive affect was measured with ten items measuring positive affect from the PANAS scale (Watson et al., 1988). Anger was adapted from Dillard and Peck (2001) based on four items (irritated, angry, annoyed, and aggravated). The statements were measured on a 5-point scale (1 = very slightly or not at all; 5 = extremely).

#### 2.2.4. Other measures

The questionnaire followed a similar approach to Martin et al. (2022) and assessed participants' experience with investing in financial assets, current ownership of cryptocurrency, and their interest in investing in cryptocurrency in case they did not own any.

#### 2.2.5. Statistical analysis

Our analysis followed the study's pre-registered protocol (https://osf.io/y3749/). Descriptive analysis and correlations were performed in SPSS version 28. The assessment of the path model was performed in SmartPLS 3 (Ringle et al., 2015), using the consistent PLS (PLSc) algorithm. PLSc is recommended when all latent constructs in the model are reflective, while estimates are comparable to those obtain from covariance-based structural equation modeling (Dijkstra & Henseler, 2015).

## 3. Results

### 3.1. Correlations

Table 2 presents intercorrelations between the variables. Correlation coefficients are positive and low to moderate, with an exception between buying intentions and attitudes where the correlation is high.

### 3.2. Path model

#### 3.2.1. Validation of measurement model and model fit

Since trait reactance was a second-order reflective construct, we followed a repeated indicators approach (Sarstedt et al., 2019) and used the latent variable scores as indicators of trait reactance. In a similar fashion we used the latent variable scores to estimate our final model. Finally, while in the former analysis the model fit criteria were below the suggested thresholds, we included a path between attitude and buying intention which improved the model fit.

Results in relation to measurement reliability and validity appear in the Supplementary Material. All latent variable composite reliabilities exceeded the commonly accepted threshold of 0.70, indicating a high level of internal consistency (Jarvis et al., 2003). Convergent validity was also met with all average variance extracted (AVE) values exceeding the threshold of 0.50 (Hair et al., 2017). We assessed discriminant validity using the heterotrait-monotrait ratio of correlations (HTMT), with all scores being below the threshold of 0.90 (Henseler et al., 2015). The Stone-Geisser $Q^2$ values obtained through the blindfolding procedures for attitude ($Q^2 = 0.278$), buying intention ($Q^2 = 0.719$), positive affect ($Q^2 = 0.029$), and anger ($Q^2 = 0.069$) were larger than zero, supporting the predictive relevance of the model (Hair et al., 2017). The standardized root mean square residual value for the saturated structural model was 0.031, which was below the threshold of 0.08 that indicates a good model fit (Hair et al., 2017).

#### 3.2.2. Structural model results

Table 3 presents the path estimates of the model. Trait reactance has a positive effect on positive affect (b = 0.18) and anger (b = 0.09). Regulation content (freedom vs. restriction) has a positive effect on positive affect (b = 0.09) and negative on anger (b = −0.26), thus the freedom message increases positive affect and decreases anger relative to the restriction message.

Further, attitude had a positive effect on buying intention (b = 0.73). The interaction effects between trait reactance and regulation content on positive affect and anger were both not significant. An examination of indirect effects (Table 4) shows that the freedom message resulted in positive affect which was associated with attitude which affected buying intention (b = 0.03). Likewise, the indirect effect of regulation content shows that the restriction message resulted in anger which was associated with attitude which affected buying intention (b = −0.02). Trait reactance was positively associated with anger (b = 0.09, Table 3) and positive affect (b = 0.18). In terms of mediation, trait reactance was associated with positive affect which in turn was associated with attitude which affected buying intention (b = 0.07, Table 4). For anger, the effect of trait reactance mediated by anger was marginally significant for buying intention (b = 0.01, $p = 0.058$, Table 4), but not for attitude ($p = 0.116$).

**Table 3**
Path estimates for main effects of the model.

| Path | Path Coefficient | *t*-test | *p*-value | CI 95% |
|---|---|---|---|---|
| Attitude → Buying Intention | 0.73 | 32.17 | 0.000 | 0.69–0.78 |
| Regulation Content → Positive Affect | 0.09 | 2.09 | 0.036 | 0.00–0.16 |
| Regulation Content → Anger | −0.26 | 7.94 | 0.000 | −0.33 to −0.19 |
| Trait Reactance × Regulation Content → Positive Affect | 0.05 | 1.19 | 0.235 | −0.04 to 0.14 |
| Trait Reactance × Regulation Content → Anger | −0.06 | 1.42 | 0.157 | −0.13 to 0.02 |
| Positive Affect → Attitude | 0.52 | 15.83 | 0.000 | 0.45–0.58 |
| Positive Affect → Buying Intention | 0.18 | 6.37 | 0.000 | 0.12–0.24 |
| Anger → Attitude | 0.12 | 2.53 | 0.011 | 0.03–0.21 |
| Anger → Buying Intention | 0.08 | 3.71 | 0.000 | 0.04–0.12 |
| Trait Reactance → Positive Affect | 0.18 | 4.14 | 0.000 | 0.09–0.26 |
| Trait Reactance → Anger | 0.09 | 2.37 | 0.018 | 0.02–0.17 |

**Table 4**
Path estimates for indirect effects of the model.

| Path | Path coefficient | t-test | p-value | CI 95% |
|---|---|---|---|---|
| Trait Reactance → Anger → Attitude → Buying Intention | 0.01 | 1.57 | 0.118 | 0.00–0.02 |
| Trait Reactance → Positive Affect → Buying Intention | 0.03 | 3.24 | 0.001 | 0.02–0.05 |
| Regulation Content → Positive Affect → Attitude | 0.05 | 2.04 | 0.041 | 0.00–0.09 |
| Regulation Content → Positive Affect → Attitude → Buying Intention | 0.03 | 2.05 | 0.041 | 0.00–0.06 |
| Regulation Content → Anger → Attitude | −0.03 | 2.32 | 0.020 | −0.06 to −0.01 |
| Trait Reactance → Anger → Attitude | 0.01 | 1.57 | 0.116 | 0.00–0.03 |
| Regulation Content → Positive Affect → Buying Intention | 0.02 | 1.98 | 0.047 | 0.00–0.03 |
| Anger → Attitude → Buying Intention | 0.09 | 2.52 | 0.012 | 0.02–0.15 |
| Trait Reactance → Positive Affect → Attitude → Buying Intention | 0.07 | 3.88 | 0.000 | 0.03–0.10 |
| Regulation Content → Anger → Attitude → Buying Intention | −0.02 | 2.31 | 0.021 | −0.04–0.00 |
| Positive Affect → Attitude → Buying Intention | 0.38 | 15.29 | 0.000 | 0.33–0.43 |
| Regulation Content → Anger → Buying Intention | −0.02 | 3.22 | 0.001 | −0.03 to −0.01 |
| Trait Reactance → Positive Affect → Attitude | 0.09 | 3.84 | 0.000 | 0.05–0.14 |
| Trait Reactance → Anger → Buying Intention | 0.01 | 1.90 | 0.058 | 0.00–0.02 |

## 4. Discussion

Currently, regulatory agencies are discussing potential regulations for people to buy and sell cryptocurrencies. These discussions include regulations that protect crypto consumers and may restrict who can invest in crypto (e.g., Harding, 2021; Kahan, 2021; Robinson, 2022; Smith, 2021). The purpose of the present research was to study the influence of trait reactance and messages of freedom-protecting or freedom-restricting regulations on crypto judgments. For mediators, we examined positive affect and negative affect (anger). The findings showed that freedom regulations about crypto resulted in more positive affect and less anger. In contrast, restriction regulations resulted in less positive affect and more anger. Thus, regulations that restrict consumer freedom to invest in crypto result in more anger than freedom-protecting messages.

Trait reactance was associated with experiencing anger. Trait reactance was also positively associated with experiencing positive affect. The limited prior research in this area suggests trait reactance is negatively associated with positive affect. Moreira et al. (2020) found in a sample of Portuguese students, that trait reactance was negatively associated with positive affect ($r = -0.16$, $p < 0.01$). Thus, our research contributes by showing how trait reactance can be associated with positive affect in the context of crypto investing regulations. Further, we found that the effect of trait reactance on crypto attitude and buying

intention was mediated by positive affect. Prior research found that the crypto purchase intentions of consumers can be mediated by positivity (Martin et al., 2022). As crypto involves peer-to-peer transactions with no government intermediary, we speculate that crypto may be viewed as a vehicle for freedom, free from government interference. This perceived freedom inherent in cryptocurrency may especially appeal to high-trait reactance people and be responsible for the positive affect from people higher in trait reactance for messages about crypto. Future research should examine how crypto is perceived by individuals who value autonomy and personal freedom. Indeed, we hope our research encourages researchers to broaden their study of trait reactance to include positive affective responses.

The present research also builds on prior research which shows the value of studying individual differences for consumer responses to cryptocurrencies. For example, Sudzina et al. (2021) found that early adopters of crypto were lower in agreeableness and higher in extraversion. Martin et al. (2022) found that the Dark Tetrad was associated with crypto judgments. Our research complements this work by studying another individual difference (trait reactance) that influences consumer cryptocurrency buying intention.

## 5. Limitations and future research

A limitation of our research is that actual buying behavior was not examined. Further, additional mediators could have been studied. Future research should examine buying behavior and other mediators that may be associated with reactance, such as conspiracy beliefs which have been shown to mediate the effects of the Dark Tetrad on cryptocurrency buying intentions (Martin et al., 2022). Our mediational model could also be augmented with cognition. For example, negative cognition could be included to measure state reactance (Dillard & Shen, 2005). Further, future research could extend our model by examining the role of positive cognitions (Garland et al., 2015; Macleod & Moore, 2000). Alternatively, researchers could examine positive emotions (e.g., gratitude) or more neutral emotions (e.g., surprise) which may be activated by freedom-protecting regulations. Other individual differences such as dispositional greed and materialism should also be studied for crypto buying intention.

## 6. Conclusion

With the crypto market worth over USD$1.7 trillion dollars, crypto trading regulations are being introduced (Kahan, 2021; Robinson, 2022). Yet there is uncertainty about what these regulations should involve and emphasize (Smith, 2021). Our research shows higher trait reactance individuals can experience anger at regulations which influences their buying intention. However, they can also experience positive affect. Across individuals, regulations that protect an individual's freedom to buy crypto generate positive affect and less anger from people than regulations which restrict crypto investment. Researchers and public policy makers should be aware that consumers do not view crypto regulations as a boring, dry issue. Instead, they generate both positive and negative feelings from consumers.

## CRediT authorship contribution statement

**Brett A.S. Martin:** Conceptualization, Writing – original draft, Writing – review & editing. **Polymeros Chrysochou:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Carolyn Strong:** Conceptualization, Writing – original draft.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.paid.2022.111659. Data and pre-registered protocol can be accessed at: https://osf.io/y3749/.

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